

Implementation Assignment 3

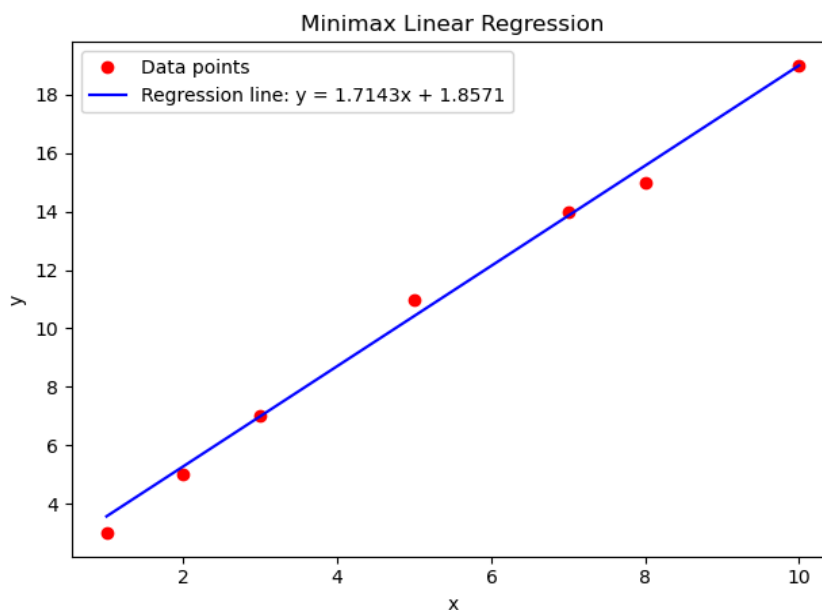
Problem 1

Given a set of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the goal is to find a line $y = ax + b$ that minimizes the maximum absolute deviation from the data. To model this as a linear program, we introduce a slack variable E which represents the worst-case deviation across all points. The constraint that $|ax_i + b - y_i| \leq E$ for all i can be linearized into two one-sided inequalities per point, giving the following formulation:

Minimize E , subject to $ax_i + b - E \leq y_i$ and $-ax_i - b - E \leq -y_i$ for all i , with $E \geq 0$ and a, b unrestricted in sign.

I used Gurobi in my implementation, which solved this instance in 4 simplex iterations in under 0.01 seconds, with an optimal objective value of 0.5714. The resulting line of best fit is $y = 1.7143x + 1.8571$, with a maximum absolute deviation of 0.5714 across all seven data points.

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Coefficient statistics:
Matrix range [1e+00, 1e+01]
Objective range [1e+00, 1e+00]
Bounds range [0e+00, 0e+00]
RHS range [3e+00, 2e+01]
Presolve time: 0.00s
Presolved: 14 rows, 3 columns, 42 nonzeros
Iteration Objective Primal Inf. Dual Inf. Time
0 handle free variables 0s
4 5.7142857e-01 0.000000e+00 0.000000e+00 0s
Solved in 4 iterations and 0.00 seconds (0.00 work units)
Optimal objective 5.714285714e-01
Optimal a: 1.7142857142857142
Optimal b: 1.8571428571428572
Optimal t (max absolute deviation): 0.5714285714285716
Plot saved as regression_plot.png
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Problem 2

To fit the temperature model to the Corvallis data, we use the same minimax approach from Problem 1, but with six free variables (x_0 through x_5) to describe the linear trend and sinusoidal components. Each daily temperature observation (d_i, T_i), we require that the model prediction $T(d_i)$ deviates from the observed temperature by no more than E . This absolute value condition is linearized into two inequality constraints per data point, with the goal to minimize E . All of x_0 through x_5 are unrestricted in sign, and E is constrained to be non-negative.

Gurobi solved this LP using a sifting method, completing 69 iterations in 0.38 seconds. The optimal maximum absolute deviation was 14.24°C, which reflects the natural day-to-day variability in temperature that a smooth sinusoidal model cannot capture. The recovered parameters were $x_0 = 8.0214$, $x_1 = 0.00010695$, $x_2 = 4.2809$, $x_3 = 8.1869$, $x_4 = -0.7906$, and $x_5 = -0.2954$.

Coefficient statistics:

Matrix range [6e-05, 2e+04]

Objective range [1e+00, 1e+00]

Bounds range [0e+00, 0e+00]

RHS range [2e-01, 3e+01]

Presolve time: 0.22s

Presolved: 7 rows, 44608 columns, 312220 nonzeros

Iteration	Objective	Primal Inf.	Dual Inf.	Time
0	7.0015891e+33	1.726236e+34	7.001589e+03	0s

Starting sifting (using simplex for sub-problems)...

Iter	Pivots	Primal Obj	Dual Obj	Time
0	19	-infinity	1.9194274e+01	0s
1	27	-0.0000000e+00	1.7548025e+01	0s
2	44	1.2912392e+01	1.6577204e+01	0s
3	64	1.4076067e+01	1.5322406e+01	0s

Sifting complete

69	1.4235540e+01	0.000000e+00	0.000000e+00	0s
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Solved in 69 iterations and 0.38 seconds (0.28 work units)

Optimal objective 1.423554030e+01

Optimal solution found:

$x_0 = 8.021419721336494$

$x_1 = 0.00010694836293243716$ (daily drift in °C)

$x_2 = 4.28089074750564$

$x_3 = 8.186857819354794$

$x_4 = -0.7906307941032016$

$x_5 = -0.2953602081034127$

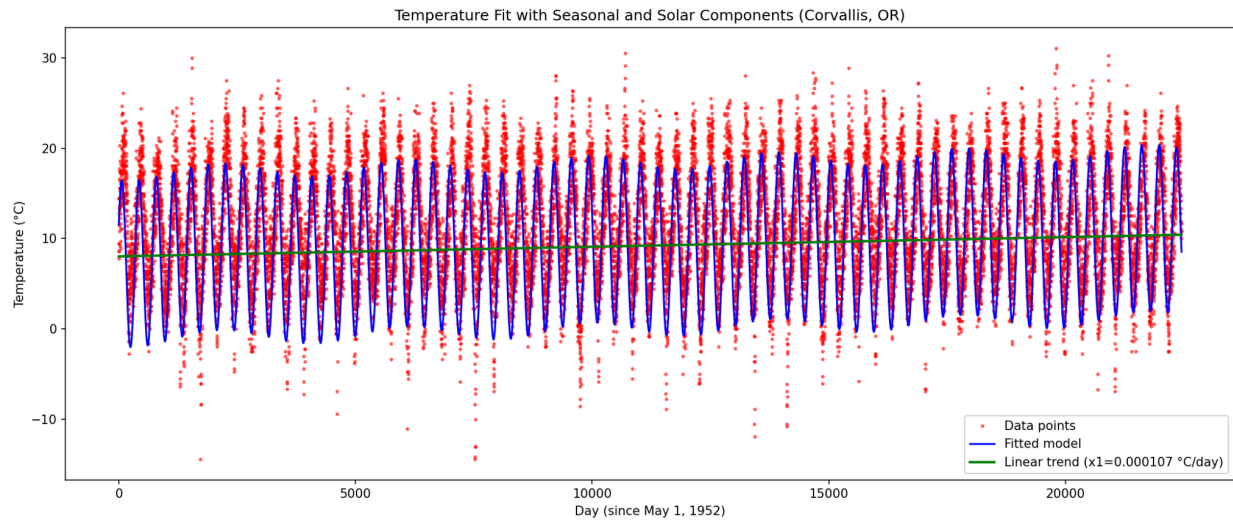
Minimum maximum absolute deviation (E) = 14.235540295367645

Estimated annual drift ($x_1 * 365.25$) = 0.03906288956107267 °C/year

Estimated drift per century ($x_1 * 365.25 * 100$) = 3.9063 °C/century

This represents a warming trend.

Plot saved as 'temperature_fit.png'



The linear drift coefficient $x_1 = 0.00010695^\circ\text{C}/\text{day}$ translates to an annual warming rate of approximately $0.039^\circ\text{C}/\text{year}$, or roughly 3.91°C per century. The model indicates that Corvallis has in fact been changing, and the data shows we are experiencing a warming trend.

Instructions for running

My submission includes only problem1.py and problem2.py, which can be run using the python interpreter. The pandas, numpy, matplotlib, and gurobipy packages must be installed. As well as a license is required to use gurobipy. I wrote my code on my linux system, where gurobipy required me to place the license file called "gurobi.lic" that you can download from their website in my home directory. If you are running this on windows, the license file is required to be in the user directory found at C:\Users\YOUR_USERNAME.